

Md Hasebul Hasan

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RESEARCH INTEREST

- My research interests lie at the intersection of **Human-AI Interaction** and **Natural Language Processing**. I aim to develop trustworthy AI systems by creating methods that diagnose when and why AI models fail and by designing mitigation strategies that reduce these failures and enable reliable, human-centered deployment.

EDUCATION

- Ph.D. in CSE, University of Texas at Arlington, Arlington, TX** [Aug 2025 – Present]
CGPA: 4.00/4.00
Full-time Ph.D. Student and Graduate Research Assistant
Supervisor: Dr. Mohammad Atiqul Islam
- M.Sc. in CSE, Bangladesh University of Engineering and Technology, Dhaka** [Jul 2022 – Aug 2025]
CGPA: 3.75/4.00
Authored three peer-reviewed paper in ICML, EACL, and COMPSAC.
Thesis: *AN EFFICIENT COMPOSITIONAL REASONING APPROACH FOR SOLVING GEOSPATIAL QUERIES USING LLMs*
- B.Sc. in CSE, Bangladesh University of Engineering and Technology, Dhaka** [Feb 2017 – May 2022]
CGPA: 3.77/4.00 (3.89 in senior year), Rank 31 out of 127
Thesis: *A MULTI-MODAL DEEP LEARNING BASED APPROACH FOR HOUSE PRICE PREDICTION.*

PUBLICATIONS

Conference Proceedings

- Dihan, M.L., Hassan, M.T., Parvez, M.T., **Hasan, M.H.**, Alam, M.A., Cheema, M.A., Ali, M.E. and Parvez, M.R. MapEval: A Map-Based Evaluation of Geo-Spatial Reasoning in Foundation Models. In *Proceedings of the Forty-Second International Conference on Machine Learning (ICML)*, 2025. [**Spotlight - Top 2.6%**]
Team collaboration: Led literature review identifying evaluation gaps; designed the task taxonomy for spatial reasoning; contributed to dataset construction, annotation, and human verification.
- Hasan, M.H.**, Dihan, M.L., Hashem, T., Ali, M.E. and Parvez, M.R. MapAgent: A Hierarchical Agent for Geospatial Reasoning with Dynamic Map Tool Integration. In Findings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2026).
First author: Led framework design, implementation, and comprehensive evaluation demonstrating state-of-the-art performance.
- Hasan, M.H.**, Khan, M.S.I., Haque, M.S., Ahmed, A.Y. and Adnan, M.A. (2024, July). GuardFL: Detection and Removal of Malicious Clients Through Majority Consensus. In *2024 IEEE 48th Annual Computers, Software, and Applications Conference (COMPSAC)* (pp. 1112–1117). IEEE Computer Society.
First author: Led research design, algorithm development, and evaluation.
- Hasan, M.H.**, Charu, K.H., Sridhar, E.P., Deb, S. and Islam, M.A. (2026). DeEscalWild: A Real-World Benchmark for Automated De-Escalation Training with SLMs. *Submitted to the Neural Information Processing Systems Evaluations and Datasets Track (Neurips '26)*.
First author: Led benchmark creation and SLM evaluation.
- Hasan, M.H.**, Jahan, M.A., Ali, M.E., Li, Y.F. and Sellis, T. (2026). A Multi-Modal Deep Learning Based Approach for House Price Prediction. *Submitted to the 18th ACM Multimedia Systems Conference (MMSys 2027), Ghent, Belgium.*
First author: Led multimodal architecture design and evaluation.

INDUSTRY EXPERIENCE

- Machine Learning Engineer, IQVIA, DHAKA

[May 2022 – Jul 2025]
- Analyzed time series data sourced from Snowflake data marts, performing comprehensive data engineering, preprocessing, and algorithm optimization
 - Developed, optimized, and deployed machine learning prediction models using PyTorch and TensorFlow within robust ML infrastructures
 - Architected multi-agent systems using LangChain for inter-agent communication and task orchestration, and fine-tuned open-source LLMs for specific applications
 - Conducted AI security red teaming to identify and mitigate potential vulnerabilities, biases, and risks in machine learning systems
 - Implemented MLOps practices for scalable model training and deployment utilizing Kubernetes, Kube-flow, AWS SageMaker, and AWS Bedrock
 - Managed codebases and developed automated CI/CD pipelines using GitLab for efficient and reliable system deployments

TEACHING EXPERIENCE

- Lecturer (Part Time), Department of CSE, BUET, DHAKA

[Nov 2022 – Jul 2024]

(Concurrent with M.Sc. studies and industry work)
- Instructed six undergraduate courses: CSE462 (Algorithm Engineering Sessional), CSE408 (Software Development Sessional), CSE216 (Database Sessional), CSE208 (Data Structures and Algorithms II Sessional), CSE206 (Digital Logic Design Sessional), and CSE102 (Structured Programming Language Sessional).
 - Supervised 30 student projects across software development, algorithms, and database design, providing mentorship on technical problem-solving and best practices
 - Designed and graded assignments, laboratory sessions, and examinations for classes of 40 – 60 students

TECHNICAL SKILLS

Programming	Python (expert), C/C++, Java, JavaScript, SQL, LaTeX
ML/AI Frameworks	PyTorch, TensorFlow, HuggingFace, LangChain, scikit-learn
Database	PostgreSQL, MySQL, MongoDB, Snowflake
Web Development	Node.js, Vue.js, React
Cloud & Tools	AWS (SageMaker, EC2, ECR, Bedrock), Docker, Kubernetes, Git

ACADEMIC GRANTS

- Research Grant for M.Sc. Thesis, DIRECTORATE OF ADVISORY, EXTENSION AND RESEARCH SERVICES, BUET, [2024]

Awarded competitive grant (1, 500 USD) for building MapAgent

AWARDS and ACHIEVEMENTS

- Platinum Award, IQVIA Impact Program, [2024]
- Silver Award, IQVIA Impact Program, [2025]
- University Merit List Scholarship (two semesters)
- University Dean's list for three consecutive years (Levels 2 to 4)

REFERENCES

Dr. Mohammad Atiqul Islam , Associate Professor

Department of Computer Science and Engineering,

University of Texas at Arlington

Relationship: PhD Supervisor

Email: mislam@uta.edu

Dr. Mohammed Eunus Ali, Associate Professor

Faculty of Information Technology,

Monash University, Melbourne, Australia

Relationship: M.Sc. Co-Supervisor and B.Sc. Thesis Supervisor

Email: Eunus.Ali@monash.edu

Dr. Tanzima Hashem, Professor

Department of Computer Science and Engineering (CSE),

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

Relationship: M.Sc. Thesis Supervisor

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